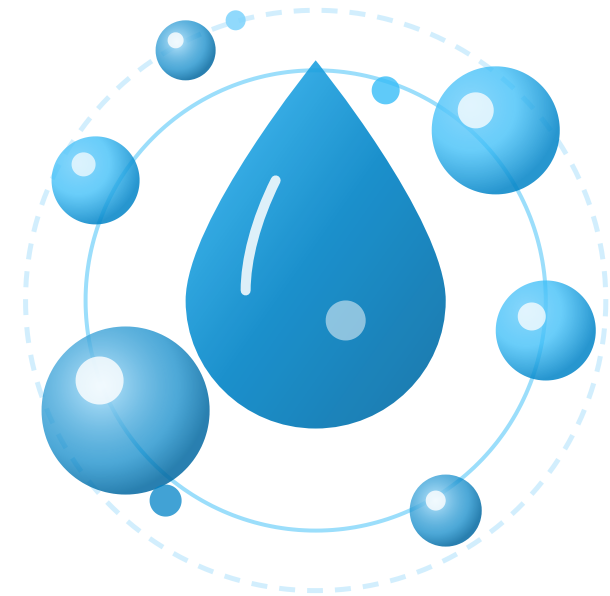


## TEAM ACE PROJECT PRESENTATION

# SMART DRAIN & HEATWAVE RESPONSE SYSTEM

An integrated urban climate management platform that uses real-time weather API forecasts and IoT sensor data to proactively address **heatwave droughts** (automated micro-irrigation) and **torrential rain** (smart pre-storm drainage) using a single rainwater storage tank.



## PROJECT TITLE

Smart Drain &amp; Heatwave System

## TEAM

Team ACE

# Extreme Summer Climate Hazards in Urban Environments

Modern cities face compound climate hazards from alternating heatwaves and flash floods.



## 1. Heatwave & Drought

- Extreme summer heat dries soil below 30% moisture.
- Urban heat island effect causes plant wilting and severe pedestrian discomfort.
- Manual watering wastes valuable municipal tap water.



## 2. Torrential Rain & Flooding

- Sudden downpours instantly exceed local storm sewer capacity.
- Rainwater tanks fill to 100% capacity and overflow into streets.
- Lack of pre-storm drainage leads to localized urban flooding.



## 3. Limits of Conventional Systems

- Existing drains operate as one-way reactive systems.
- Stormwater is dumped into sewers instead of being stored.
- No integration between flood prevention and heatwave irrigation.



## The Need for an Integrated Solution

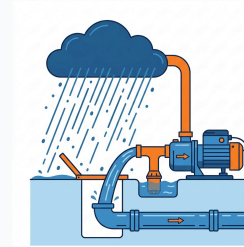
Urban infrastructure requires a smart system capable of automatically switching between **storing rainwater for drought irrigation** and **pre-draining tanks before severe storms land**.

# One Tank, Two Jobs: Bi-Directional Climate Management



## JOB 1: Heatwave Micro-Irrigation (P1 Pump)

- **Trigger Condition:** Clear weather (no rain forecast), Soil moisture < 30%, Temp  $\geq 30^{\circ}\text{C}$ , and Tank storage > 20%.
- **Control Action:** Activates **P1 Irrigation Pump** to deliver a 12-second pulse (~0.34 L) to the crop box.
- **Benefit:** Recycles stored rainwater to preserve urban crops without wasting tap water.



## JOB 2: Pre-Storm Flood Drainage (P2 Pump)

- **Trigger Condition:** KMA Open API receives a Heavy Rain Forecast AND Tank storage level > 20%.
- **Control Action:** Activates **P2 Pre-drain Pump** to empty the tank down to 10% before downpour hits.
- **Benefit:** Creates an empty storage buffer in advance, preventing urban sewer overload.

## The Rainwater Circularity Loop

The torrential downpour that threatens street flooding is stored into the tank, becoming the exact water source that cools the city during the next heatwave.

# Dumb Edge Node, Smart Hub System Architecture

## 1. ESP32 Edge Node

- **Sensors:** Capacitive Soil Moisture (ADC1), DHT22 Temp/Humidity, HC-SR04 Ultrasonic Tank Level.
- **Actuators:** 2-Channel 5V Relay controlling P1 & P2 Submersible Pumps.
- **Role:** Reads telemetry every 2 s, reports via MQTT, executes commands with 15 s safety watchdog.

## 2. Raspberry Pi Main Hub

- **Mosquitto MQTT Broker:** Handles real-time JSON messaging.
- **decision.py:** Central decision engine evaluating state logic every 5 s.
- **FastAPI & SQLite:** Serves system dashboard and records event telemetry logs.

## 3. KMA Forecast API

- **Open API Polling:** Fetches short-term forecast every 10 min.
- **Grid Conversion:** Converts lat/lon to Lambert conformal conic grid.
- **Fail-Safe:** API outages default to "last valid forecast" then "no rain", preventing erroneous drainage.

### Telemetry Payload (ESP32 → Pi, 2 s)

```
{ "ts": 1786342072.4, "node": "ace-esp32", "soil_moisture_pct": 27.4, "ambient_temp_c": 34.2, "tank_level_pct": 43.1 }
```

### Command Payload (Pi → ESP32, 5 s)

```
{ "mode": "GREENERY_IRRIGATION", "pump": true, "drain": false, "reason": "moisture 27%, 34.2C", "ttl_s": 15.0 }
```

# System Decision Logic

## CORE DECISION ENGINE

## 5-Second Real-Time Decision Algorithm

Evaluates sensor data and weather forecast every 5 seconds to automatically toggle between pre-storm flood protection and heatwave micro-irrigation.

### ⌚ Real-Time Sensor & Weather Polling

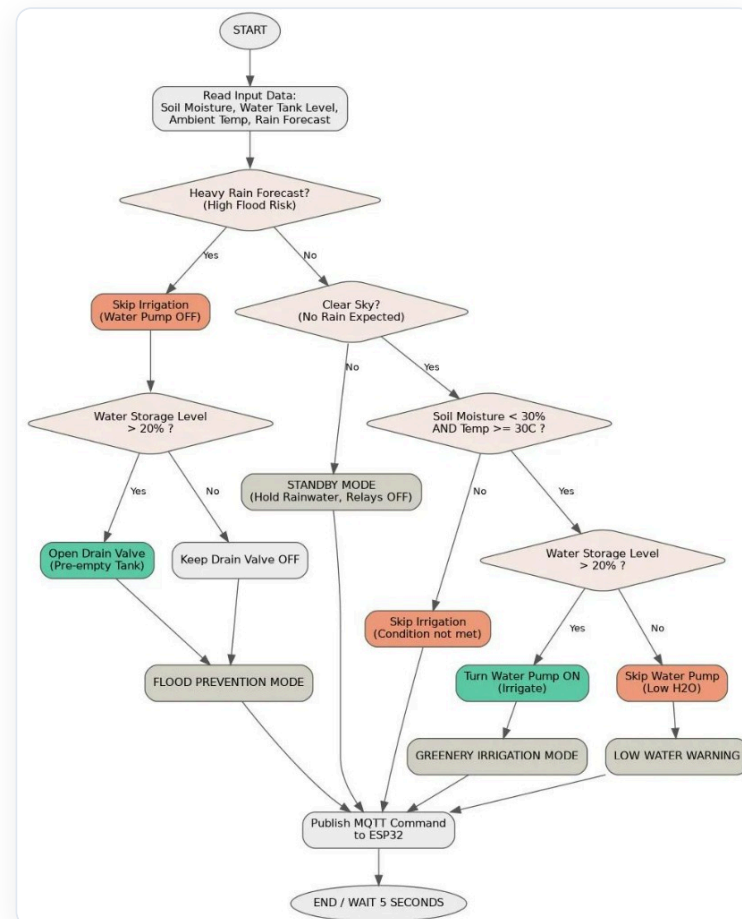
Polls KMA forecast API, soil moisture, ambient temp, and water tank storage level every 5-second cycle.

### ☔ Heavy Rain Priority Override

Heavy rain forecast triggers instant pre-drainage via P2 pump, clearing tank storage capacity before storm hits.

### 🌿 Smart Micro-Irrigation

Doses a 0.34 L pulse via P1 pump during hot ( $\geq 30^{\circ}\text{C}$ ) and dry ( $< 30\%$  moisture) conditions when stored water is available.



# Conventional Systems vs. Team ACE SDS Smart Platform

| Feature              | Conventional Municipal Drainage          | ACE SDS Smart Platform                                      |
|----------------------|------------------------------------------|-------------------------------------------------------------|
| Primary Goal         | Flood prevention only                    | Flood + Heatwave + Drought integrated control               |
| Rainwater Processing | Store then discharge into sewer (wasted) | Filtered → Stored → 100% Recycled for Irrigation & Mist     |
| Water Flow Mechanism | One-way gravity discharge                | Bi-directional smart routing (P1 Irrigation & P2 Pre-drain) |
| Decision Timing      | Reactive response after heavy rain       | Proactive 1-hour KMA forecast API poller                    |
| Application Scope    | Large public civil infrastructure        | Roofs, planter boxes, smart shade micro-zones               |

 **Smart Shade Cooling Fog Integration & Innovations**

 **Rainwater First**  
Prioritizes stored rainwater over municipal tap water.

 **Multi-Height Spray**  
Nozzle heights for children (1.2m) & adults (1.8m).

 **IoT Climate Control**  
Lowers felt ambient temperature by 3~5°C.

 **3-Tier Filtration**  
Topsoil, fabric filter, and gravel prevent clogging.



LIVE PROTOTYPE DEMONSTRATION

# "Let's Watch the Physical Prototype Demonstration Video"

Demonstrating the 1:1 physical hardware prototype — featuring real-time water pump switching for P1 (Heatwave Irrigation) and P2 (Flood Pre-drainage) based on live forecast API signals and sensor telemetry.

 1. Heatwave Micro-Irrigation (P1 Pulse) Action

 2. Pre-Storm Smart Flood Drainage (P2) Action

## 07 — CONCLUSION & Q&A

# SMART DRAIN & HEATWAVE RESPONSE SYSTEM

Thank you for your time. Ready for any questions.

Control Logic  
**100% Verified**

Simulation Unit Tests  
**15 / 15 PASS**

Weather API Grid  
**KMA Forecast Validated**

## Question & Answer (Q&A)

Team ACE · 2026